Enrollment No.: _____

Darshan UNIVERSITY
योगः कर्मसु कौशलम्

Darshan Institute of Computer Application
MCA | Sem-2 | Winter-2023

Course Code	:	2105CS202	**Date**	:	28-11-2023
Course Name	:	Python for Data Science	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** Explain data science pipeline in detail. **4**

(B) Explain python's role in data science. **3**

OR

Explain functional and imperative coding styles in python.

(C) Explain python ecosystem for data science. **7**

OR

Explain %run, %time, %timeit and %reset magic commands with example.

Q.2 **(A)** Explain database connectively with MySQL database with a simple example. **4**

(B) Explain sort() function of numpy with an example. **3**

OR

Explain conditional selection in a numpy array.

(C) Explain arange(), zeros(), ones(), eye() and linspace() functions with example. **7**

OR

Explain xs() and groupby() functions of pandas with example.

Q.3 **(A)** Explain conditional selection on DataFrame. **4**

(B) Differentiate join and merge. **3**

OR

Explain multi-index DataFrame with an example.

(C) Consider following Restaurant_order.csv **7**

order_id	quantity	item_name	choice_description	item_price

Using pandas perform the following tasks.

1) Import the dataset from 'http://abc.com/restaurant_order.csv' and assign it to a variable called 'rsorder'
2) Display first 25 entries of DataFrame.
3) How is the dataset indexed?

4) What is the number of observations in the dataset?
5) Which was the most-ordered item?
6) What was the most ordered item in the choice_description column?
7) How many items were ordered in total?

OR

Consider following Salaries.csv

Id	EmployeeName	JobTitle	BasePay	TotalPay	Year

Using pandas perform the following tasks.

1) Read Salaries.csv as a DataFrame called 'sal'.
2) Display last 25 entries of DataFrame.
3) What is the average BasePay?
4) Print the name of all the columns.
5) What job title does an "XYZ" employee hold?
6) What is the name of highest paid person?
7) Summarize the DataFrame.

Q.4 **(A)** Explain different types of line appearances available in matplotlib. **4**

(B) Explain Axis, Ticks and Grid. **3**

OR

Explain Networkx library with directed graph.

(C) Write a python program to implement LMplot and heatmap with random data. **7**

OR

Write a python program to implement countplot and boxplot with titanic dataset.

Q.5 **(A)** Differentiate Supervised Learning and Unsupervised Learning. **4**

(B) Explain confusion matrix with an example. **3**

OR

Explain bag of words model.

(C) What is regression? Explain logistic regression with an example. **7**

OR

What is clustering? Explain K-Mean with an example.

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Table showing the Bloom's Taxonomy Level and Course Outcome

Question		Bloom's Taxonomy Level *	Course Outcome**
Q.1	(A)	U	CO1
	(B)	U	CO1
	(C)	U	CO1
Q.2	(A)	U	CO2
	(B)	U	CO2
	(C)	U	CO2
Q.3	(A)	U	CO2
	(B)	U	CO2
	(C)	A	CO2
Q.4	(A)	U	CO3
	(B)	U	CO3
	(C)	A	CO3
Q.5	(A)	U	CO4
	(B)	U	CO4
	(C)	U	CO4

* **Bloom's Taxonomy Level**

R - Remembrance, **U** - Understanding, **A** - Application, **N** - Analyze, **E** - Evaluate, **C** – Create

** **Course Outcome**

At the end of the course, student will be able to:

CO1 : describe data science using python tools.

CO2 : experiment with Beautiful Soup, NumPy and Pandas library.

CO3 : sketch graphs using matplotlib library.

CO4 : analyze the supervised and unsupervised learning using scikit-learn.